



AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Project Domain

[Excel / Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



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Employee Payroll Data Cleaning and Financial Analysis

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ABSTRACT OF THE PROJECT

The project titled “Employee Payroll Data Cleaning and Financial Analysis” focuses on transforming raw payroll data into meaningful financial insights using Microsoft Excel and Artificial Intelligence tools. Payroll systems are essential for managing employee salaries, bonuses, tax deductions, and payment records in organizations. However, payroll datasets often contain inconsistencies such as missing values, duplicate records, incorrect salary entries, invalid tax calculations, and inconsistent date formats, which affect the accuracy of financial reporting and decision-making.

The main objective of this project is to clean and preprocess payroll data to improve data accuracy and reliability. The project involves identifying and correcting errors, handling missing values, removing duplicate records, standardizing formats, and validating salary-related calculations using Excel tools and formulas.

After the cleaning process, financial analysis was performed using Excel formulas such as SUM, AVERAGE, IF, SUMIF, COUNTIF, and VLOOKUP along with Pivot Tables for department-wise salary analysis and payroll expense calculation. Various charts and dashboards were created to visualize employee salary distribution, bonus analysis, tax deductions, and department-wise payroll expenses.

Additionally, Artificial Intelligence tools such as ChatGPT and Microsoft Copilot were integrated into the project to generate automated summaries, payroll insights, and recommendations. AI-generated narratives helped simplify data interpretation and improve reporting efficiency.

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CHAPTER 1

Introduction

1.1 Problem Statement

Organizations generate large amounts of payroll data related to employee salaries, bonuses, tax deductions, and payment records. However, payroll datasets often contain inconsistencies such as missing values, duplicate records, incorrect salary calculations, invalid tax entries, and inconsistent date formats. These issues reduce data accuracy and make payroll management difficult.

Managing payroll data manually becomes time-consuming and increases the possibility of human errors, especially when organizations handle large employee datasets. Incorrect payroll records may lead to salary miscalculations, financial discrepancies, and poor decision-making.

This project aims to address these challenges by:

- Cleaning and preprocessing employee payroll data
- Performing financial analysis using Excel formulas and Pivot Tables
- Creating charts and dashboards for visualization
- Integrating AI tools to generate automated insights and summaries

The main goal of the project is to transform raw payroll data into a clean, structured, and meaningful format that improves payroll accuracy and supports better financial decision-making.

1.2 Motivation

The main motivation behind this project is to improve payroll data accuracy and simplify financial analysis using Excel and Artificial Intelligence tools.

The project also helps in understanding how AI tools such as ChatGPT and Microsoft Copilot can generate insights and summaries from payroll data efficiently.

1.3 Objectives

- To collect and analyze a payroll dataset
- To remove duplicate records and missing values
- To correct incorrect salary and tax entries
- To perform financial analysis using Excel formulas and Pivot Tables

1.4 Scope of the Project

This project focuses on employee payroll data cleaning and financial analysis using Microsoft Excel and AI tools. The project includes cleaning payroll data, analysing salary information, creating visual reports, and generating AI-assisted insights.

Limitations (Out of Scope):-

- The project uses a static dataset
- Real-time payroll integration is not included
- Advanced machine learning models are not implemented

CHAPTER 2

Literature Survey

A literature survey helps in understanding existing tools, techniques, and technologies related to the project. In the field of data analytics, Microsoft Excel and Business Intelligence tools are widely used for data cleaning, analysis, and visualization.

Many organizations use payroll systems to manage employee salary records, tax deductions, and financial reporting. However, payroll datasets often contain errors and inconsistencies that affect data accuracy. Data cleaning techniques help in improving the quality of payroll information and making analysis more reliable.

Artificial Intelligence tools such as ChatGPT and Microsoft Copilot are now being used to generate automated insights, summaries, and recommendations from datasets. These tools reduce manual effort and improve reporting efficiency.

2.1 Existing Work and Research

Previous studies and business practices mainly focus on payroll management systems, salary processing, financial reporting, and employee compensation analysis using Excel and Business Intelligence tools. Many organizations use payroll software and spreadsheet-based systems to maintain employee salary records and calculate payroll expenses.

Existing systems commonly use formulas, Pivot Tables, and charts for analysing salary distribution, tax deductions, and department-wise payroll expenses. Dashboards are also used to present payroll information visually for easier understanding and reporting.

Although many payroll systems provide reporting and visualization features, AI-based insight generation and automated explanations are still limited in most organizations. Many systems still depend on manual interpretation of payroll data.

2.2 Tools and Technologies Review

The following tools and technologies are used in this project:

Microsoft Excel

Microsoft Excel is used for data cleaning, salary calculations, Pivot Tables, formulas, and financial analysis. It helps in removing duplicate records, correcting errors, handling missing values, and analysing payroll data efficiently.

Microsoft Power BI

Power BI is used for creating dashboards, charts, and visual reports. It helps in visualizing department-wise salary distribution, payroll expenses, and employee compensation trends.

ChatGPT / Microsoft Copilot

AI tools such as ChatGPT and Microsoft Copilot are used to generate summaries, payroll insights, and recommendations from the analysed data. These tools improve reporting efficiency and simplify data interpretation.

2.3 Gaps Identified

- Manual payroll analysis requires more time and effort
- Existing payroll systems provide limited automated insights
- Large payroll datasets are difficult to manage manually
- Many systems lack AI integration for intelligent reporting and analysis
- Manual verification increases the chances of payroll errors and inconsistencies

CHAPTER 3

Proposed Methodology

3.1 System Design

The proposed system is designed to clean and analyse employee payroll data using Microsoft Excel and Artificial Intelligence tools. The project follows a structured process from data collection to final analysis and visualization.

The workflow of the project includes the following steps:

- **Data Collection** : Collecting the payroll dataset containing employee salary details, bonuses, tax deductions, and payment records.
- **Data Cleaning** : Identifying and correcting errors such as missing values, duplicate records, incorrect salary entries, and inconsistent date formats.
- **Financial Analysis** : Performing salary analysis using Excel formulas, Pivot Tables, and calculations.
- **Visualization** : Creating charts and dashboards to represent payroll insights visually.
- **AI Integration** : Using AI tools such as ChatGPT and Microsoft Copilot to generate summaries, insights, and recommendations.

The system helps transform raw payroll data into a clean and structured format that improves reporting and decision-making.

3.2 Modules / Components Used

The project is divided into the following modules:

- Data Collection Module : Used for importing and understanding the payroll dataset containing employee salary information.
- Data Cleaning Module : Used for removing duplicates, handling missing values, correcting invalid entries, and standardizing data formats.
- Financial Analysis Module : Used for salary calculations, tax analysis, bonus analysis, and department-wise payroll analysis using Excel formulas and Pivot Tables.
- Visualization Module : Used for creating charts, graphs, and dashboards to represent payroll information visually.
- AI Integration Module : Used for generating AI-based summaries, payroll insights, and recommendations using ChatGPT and Microsoft Copilot.

3.3 Data Flow Diagram (DFD)

Represents the overall system flow:-

PAYROLL DATASET → DATA CLEANING → FINANCIAL ANALYSIS → DASHBOARD & AI INSIGHTS

DFD Level 0 — Context Diagram:-

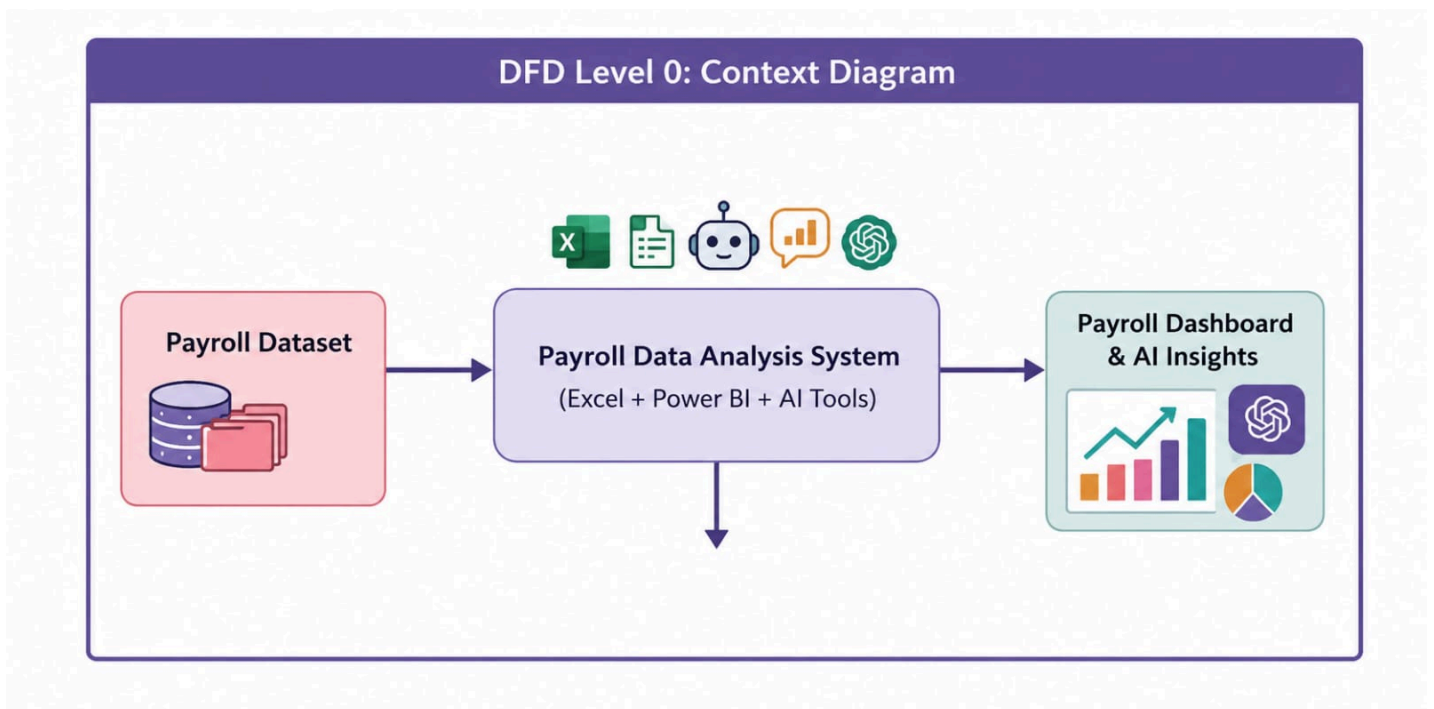


Figure : 1 DFD Employee Payroll Flow

The system receives payroll data as input, processes the data through cleaning and analysis techniques, and produces dashboards and AI-generated insights as output.

DFD Level 1 — Detailed Process Flow

Breakdown of processes:

1. Data Collection
2. Data Cleaning and Validation
3. Salary and Tax Analysis
4. Dashboard and Chart Creation
5. AI-Based Insight Generation

3.4 Advantages

- Reduces manual effort in payroll analysis
- Improves payroll data accuracy and consistency
- Helps identify salary and tax-related insights easily
- Provides visual dashboards for better understanding
- Integrates AI tools for automated reporting and summaries
- Improves decision-making and reporting efficiency

3.5 Requirement Specification Hardware Requirements

Component	Specification
Processor	Intel Corei3orhigher (i5/i7 recommended)
RAM	Minimum4GB(8GB recommended)
Storage	Minimum50GBfree disk space
Display	1366 x 768resolution or higher
Internet	RequiredforCopilot / Chatbot access

3.6 Software Requirements

Software	Details
OperatingSystem	Windows 10 / 11 or macOS
MicrosoftExcel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BIDesktop	Latest version (free download from Microsoft)
MicrosoftCopilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
WebBrowser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

4.1 Data Collection and Dataset Description

The project uses an employee payroll dataset containing employee details such as employee ID, department, salary, bonus, tax deduction, and payment information. The dataset was collected and maintained in Microsoft Excel for analysis and reporting purposes.

Employee ID	Employee Name	Gender	Age	Department	Designation	Joining Date	Basic Salary	Allowances	Bonus	Tax Deduction	Net Salary	Performance	Payment Date
EMP001	Bryan Dav	Male		Ops	Operation	22-05-202	81395	10834	11533	11956.27	91805.73	85	5 06-05-2026
EMP002	Michael F	Male	29	sales	Sales Mar	2019/01/3	81017	11770	4228	13469.62	83545.38		2 04/11/2026
EMP003	Daniel Da	FEMALE		Finance	Accountar	11/20/202	59,632	13905	3102	4064.77	72574.23	100	04/15/2026
EMP004	Glenn Alvi	FEMALE		H.R	HR Execu	04-03-202	64657	14838	4003	12119.73	71378.27		4 2026/05/03
EMP005	Shawn Sa	FEMALE		MKT	SEO Anal	13-11-202	46087	8134	6419	5458.15	55181.85		5 04/22/2026
EMP007	Evan Daw	male		Ops	Operation	2021/11/1	75035	6209	8355	11837.25	77761.75		1 2026/04/06
EMP008	Jenna Dav	Female		hr	HR Manag	2021/05/0	72709	3725	5085	7230.48	74288.52	92	5 2026/05/07
EMP009	Briana Hai	Male		sales	Sales Mar	09-03-202	78018	15647	9903	14739.1	88828.9		3 2026/03/31
EMP010	Daniel Thc	Male		I.T	System Ar	02/23/202	77529		12756	14760.75	91380.25	75	2026/04/15
EMP011	Brian Smit	male	24	Operation	Operation	28-12-202	28,693	5651	1141	5369.16	30115.84		4 2026/03/26
EMP012	Mary Burk	male	24	it	System Ar	2019/03/1	63295	11730	9747	10067.65	74704.35		4 2026/03/20
EMP013	Gary Garc	FEMALE		MKT	SEO Anal	06/27/202	889317	11548	11378	12027.22	100215.8		5 2026/04/06
EMP014	Robert He	male		HR	HR Execu	10/30/202	83,934	15782	9372	17638.18	91449.82	86	3 2026/05/01
EMP015	Linda Gral	Male		MKT	Marketing	2017/04/1	889276	8525	14455	15941.18	96314.82		5 2026/03/22
EMP016	Ian Robbir	FEMALE		Operation	Operation	04-05-202	67423	11560	4074	8022.36	75034.64		3 27-04-2026
EMP017	Danielle D	Female	26	SALES	Sales Mar	06/08/202	69,289	12836	1904	12218.02	71810.98	74	4 2026/04/12
EMP018	Joshua C	FEMALE		I.T	Software	11/15/202	42767	11681	8620	10328.28	52739.72		3 04/14/2026
EMP019	Tara Price	male	54	hr	HR Execu	08/25/202	84,989	8585	12711	12772.02	93512.98	98	5 30-03-2026
EMP020	Austin Rey	Male		FINANCE	Accountar	06/18/202	51611	16187	14796	6910.58	75683.42		2 19-04-2026

Figure : 2 Employee Payroll Dataset (Uncleaned Version)

In the Figure : 2, The dataset initially contained inconsistencies such as duplicate records, missing values, incorrect salary entries, and inconsistent formatting. These issues affected the accuracy of payroll analysis and required proper data cleaning before further processing.

4.2 Data Cleaning Process

Data cleaning was performed using Microsoft Excel to improve the quality and accuracy of the payroll dataset. The cleaning process included identifying duplicate records, correcting missing values, standardizing date formats, and verifying salary and tax calculations.

Employee ID	Employee Name	Gender	Age	Department	Description	Date	Salary	Allowances	Taxes	Deductions	Net Salary	Attendance Percentage	Performance Rating	Review Date
EMP001	Bryan Davis	Male	28	Operations	Operations Executive	03-08-2019	\$ 81,395.00	\$ 10,934.00	\$ 11,533.00	\$ 11,655.27	\$ 61,806.73	85%	5.0	24-11-2026
EMP002	Michael Farley	Male	29	Sales	Sales Manager	01-01-2022	\$ 81,017.00	\$ 11,770.00	\$ 4,228.00	\$ 13,459.82	\$ 63,546.38	92%	2.0	27-02-2026
EMP003	Daniel Day	Female	31	Finance	Accountant	24-12-2019	\$ 59,832.00	\$ 13,905.00	\$ 3,102.00	\$ 4,064.77	\$ 72,574.23	78%	3.0	13-01-2026
EMP004	Glenn Alvarez	Female	26	H.R.	HR Executive	27-07-2023	\$ 64,657.00	\$ 14,838.00	\$ 4,003.00	\$ 12,119.73	\$ 71,378.27	88%	4.0	21-05-2026
EMP005	Shawn Sandover	Female	24	Marketing	SEO Analyst	27-12-2021	\$ 45,087.00	\$ 8,134.00	\$ 5,419.00	\$ 5,455.15	\$ 55,181.85	85%	5.0	06-05-2026
EMP007	Jenna David MD	Female	39	H.R.	HR Manager	17-03-2020	\$ 72,709.00	\$ 3,725.00	\$ 5,585.00	\$ 7,230.45	\$ 74,288.52	91%	5.0	25-04-2026
EMP008	Briana Hamilton	Male	28	Sales	Sales Manager	06-06-2019	\$ 78,018.00	\$ 15,647.00	\$ 9,903.00	\$ 14,739.10	\$ 88,828.90	90%	3.0	13-03-2026
EMP009	Daniel Thompson	Male	40	IT	System Analyst	02-04-2023	\$ 77,529.00	\$ 8,134.00	\$ 12,755.00	\$ 14,760.75	\$ 91,380.25	78%	3.0	22-02-2026
EMP010	Brian Smith	Male	24	Operations	Operations Executive	05-01-2025	\$ 28,593.00	\$ 5,951.00	\$ 1,141.00	\$ 5,389.15	\$ 30,115.84	84%	4.0	13-12-2026
EMP011	Mary Burke	Male	24	IT	System Analyst	22-04-2025	\$ 63,295.00	\$ 11,730.00	\$ 9,747.00	\$ 10,067.65	\$ 74,704.35	97%	4.0	07-10-2026
EMP012	Gary Garcia	Female	23	Marketing	SEO Analyst	22-09-2022	\$ 89,317.00	\$ 11,548.00	\$ 11,378.00	\$ 12,027.22	\$ 100,215.78	89%	5.0	14-02-2026
EMP013	Robert Hernandez	Male	29	H.R.	HR Executive	27-10-2022	\$ 83,934.00	\$ 15,782.00	\$ 9,372.00	\$ 17,638.18	\$ 91,449.82	93%	3.0	30-10-2026
EMP015	Linda Graham	Male	38	Operations	Operations Executive	01-08-2019	\$ 57,423.00	\$ 11,580.00	\$ 4,074.00	\$ 5,022.36	\$ 75,034.64	74%	3.0	05-08-2026
EMP016	Jan Robbons	Female	35	Finance	Accountant	15-10-2020	\$ 62,597.00	\$ 5,830.00	\$ 1,448.00	\$ 11,080.28	\$ 78,792.72	86%	3.0	17-01-2026
EMP017	Danielle Doyle	Female	26	Sales	Sales Manager	07-07-2020	\$ 69,289.00	\$ 12,836.00	\$ 1,904.00	\$ 12,218.02	\$ 71,810.98	91%	4.0	16-01-2026
EMP018	Joshua Campbell	Female	21	IT	Software Engineer	13-10-2022	\$ 42,767.00	\$ 11,081.00	\$ 8,620.00	\$ 10,328.28	\$ 62,739.72	79%	3.0	17-02-2026
EMP019	Tara Price	Male	54	H.R.	HR Executive	16-04-2025	\$ 84,999.00	\$ 8,585.00	\$ 12,711.00	\$ 12,772.02	\$ 63,512.98	83%	5.0	22-04-2026
EMP020	Austin Reyes DDS	Male	23	Finance	Accountant	29-09-2022	\$ 61,811.00	\$ 16,157.00	\$ 14,790.00	\$ 6,910.55	\$ 75,655.42	96%	2.0	30-04-2026
EMP022	Mary Clark	Male	52	Operations	Operations Executive	01-10-2021	\$ 33,009.00	\$ 7,181.00	\$ 3,044.00	\$ 5,846.94	\$ 37,387.09	87%	5.0	16-09-2026
EMP023	Barbara Glover	Female	33	Finance	Finance Analyst	01-11-2020	\$ 73,329.00	\$ 15,846.00	\$ 5,815.00	\$ 16,374.99	\$ 78,015.01	80%	3.0	05-11-2026
EMP026	Andrew Garcia	Female	41	H.R.	HR Manager	07-01-2019	\$ 81,821.00	\$ 5,237.00	\$ 13,824.00	\$ 13,024.42	\$ 88,887.58	94%	5.0	14-01-2026
EMP028	Nicole Ford	Female	37	Finance	Finance Analyst	22-11-2023	\$ 72,018.00	\$ 8,134.00	\$ 5,850.00	\$ 6,053.86	\$ 87,067.14	82%	1.0	15-10-2026
EMP027	Brian Cook	Male	22	Finance	Accountant	25-12-2019	\$ 77,180.00	\$ 7,100.00	\$ 4,590.00	\$ 6,831.51	\$ 81,814.49	96%	3.0	12-04-2026
EMP028	Vanessa Bryant	Female	38	H.R.	HR Manager	13-09-2025	\$ 66,096.00	\$ 10,315.00	\$ 1,262.00	\$ 12,612.29	\$ 65,090.71	77%	3.0	29-11-2026
EMP029	Jason Marks	Male	39	IT	Software Engineer	27-03-2023	\$ 45,857.00	\$ 9,603.00	\$ 9,518.00	\$ 6,833.84	\$ 65,144.16	85%	1.0	26-12-2026
EMP030	David Manning	Female	32	IT	System Analyst	14-10-2019	\$ 90,251.00	\$ 16,067.00	\$ 10,319.00	\$ 20,584.99	\$ 65,662.31	92%	2.0	07-10-2026
EMP031	Jeremy Morris	Male	32	Operations	Operations Manager	28-01-2019	\$ 73,590.00	\$ 5,497.00	\$ 1,841.00	\$ 6,263.19	\$ 74,554.82	75%	1.0	03-08-2026
EMP032	Elizabeth Harper	Female	56	Sales	Sales Manager	16-07-2022	\$ 73,820.00	\$ 7,386.00	\$ 5,922.00	\$ 13,392.87	\$ 84,714.13	98%	5.0	23-04-2026
EMP033	Kellie Delacruz	Female	31	IT	System Analyst	11-01-2024	\$ 28,857.00	\$ 5,593.00	\$ 6,440.00	\$ 6,005.74	\$ 32,864.26	95%	3.0	18-08-2026
EMP034	Trevor Yang	Female	34	H.R.	HR Manager	22-02-2020	\$ 73,768.00	\$ 11,682.00	\$ 1,238.00	\$ 5,834.63	\$ 80,833.37	81%	4.0	29-10-2026
EMP035	Alexis Neal	Female	40	Sales	Sales Executive	29-06-2019	\$ 35,273.00	\$ 11,333.00	\$ 11,352.00	\$ 6,421.42	\$ 58,538.58	90%	3.0	23-05-2026
EMP036	Michael Williams	Female	29	Sales	Sales Manager	15-01-2020	\$ 62,436.00	\$ 9,934.00	\$ 2,038.00	\$ 11,179.73	\$ 71,218.27	75%	5.0	04-01-2026
EMP037	Lawrence Griffith	Female	29	H.R.	HR Manager	05-08-2020	\$ 77,738.00	\$ 12,645.00	\$ 9,295.00	\$ 17,255.98	\$ 82,418.02	84%	3.0	23-03-2026
EMP038	Amy Wolfe	Male	20	Sales	Sales Executive	01-06-2020	\$ 64,494.00	\$ 9,125.00	\$ 10,202.00	\$ 15,322.02	\$ 88,498.98	97%	5.0	24-12-2026
EMP039	Donna Smith	Female	40	Finance	Finance Executive	18-03-2015	\$ 57,660.00	\$ 5,261.00	\$ 4,000.00	\$ 6,180.10	\$ 68,071.77	90%	4.0	01-08-2026

Figure : 3 Cleaned Version Of Dataset

In Figure : 3, Excel features such as sorting, filtering, conditional formatting, and formulas were used to detect and correct errors in the dataset. After cleaning, the payroll data became more organized, accurate, and ready for analysis.

4.3 Financial Analysis

Financial analysis was performed using Excel formulas and Pivot Tables to generate useful payroll insights. Salary distribution, bonus allocation, department-wise payroll expenses, and tax deductions were analysed using different calculations and summary reports.

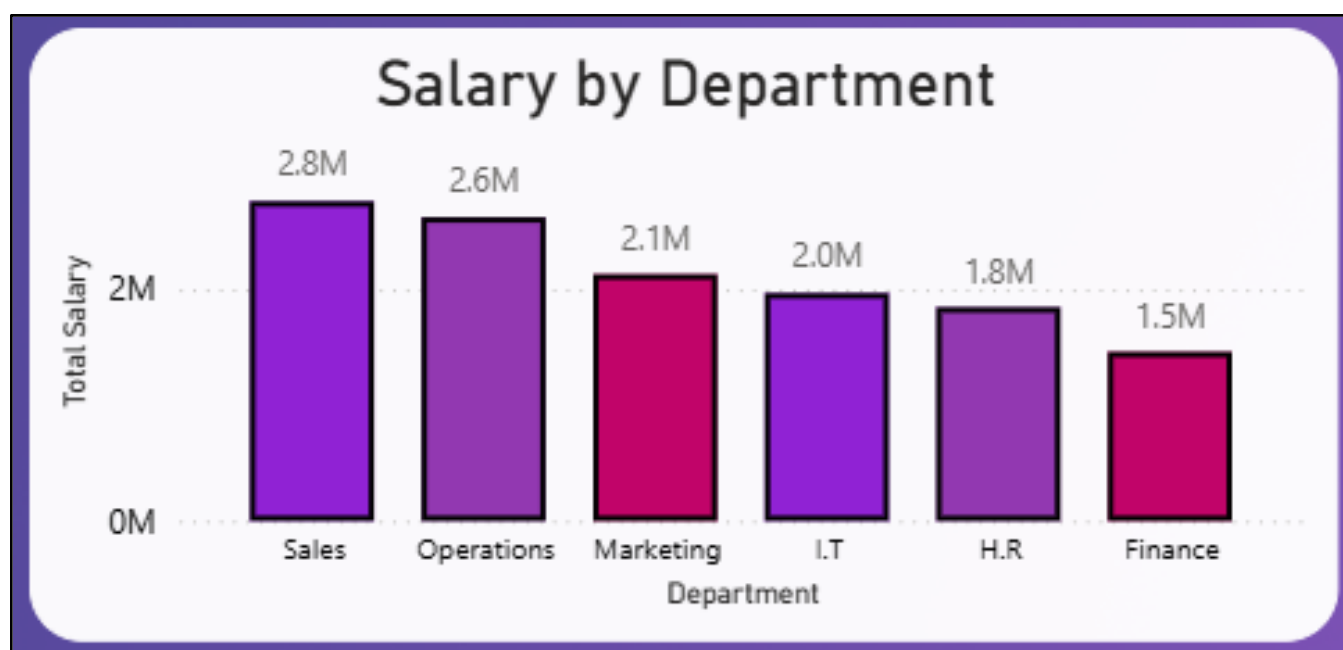


Figure : 4 Total Nat Salary Of Department

In Figure : 4, Formulas such as SUM, AVERAGE, COUNTIF, and IF statements were used to perform payroll-related calculations. The analysis helped in understanding employee compensation patterns and overall payroll expenses more effectively in a visual bar chart and pie chart format.

4.4 Visualization and Dashboard Results

Charts, graphs, and dashboards were created to visually represent payroll data and financial insights. Bar charts, pie charts, and Pivot Charts were used to display salary distribution, department-wise payroll expenses, and bonus analysis.

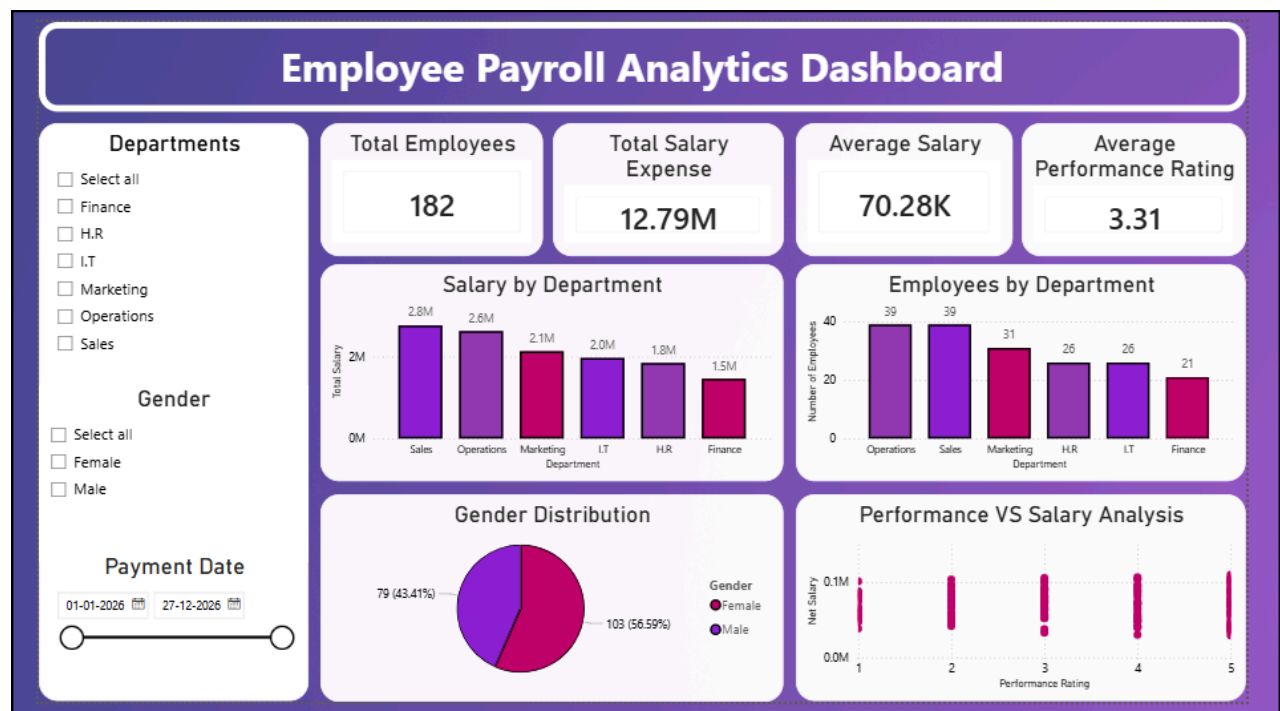


Figure : 5 Employee Payroll Dashboard

In Figure : 5, The dashboard provided a clear visual summary of payroll information and helped in understanding trends and comparisons more easily. Visualization improved the readability and presentation of payroll data.

4.5 AI Tool Integration Results

AI tools such as ChatGPT and Microsoft Copilot were used to enhance analysis.

Tool Used:-

Open AI ChatGPT / Microsoft Copilot

Sample Prompts:-

- “Summarize department-wise payroll expenses”
- “Identify trends in employee salary and tax deductions”
- “Provide recommendations based on payroll analysis”

AI Output:-

- Generated summaries of payroll trends
- Identified departments with highest salary expenses
- Suggested improvements for payroll accuracy and financial management

Impact of AI Integration:-

- Reduced manual effort in analysis
- Faster report generation
- Improved understanding of data through natural language insights

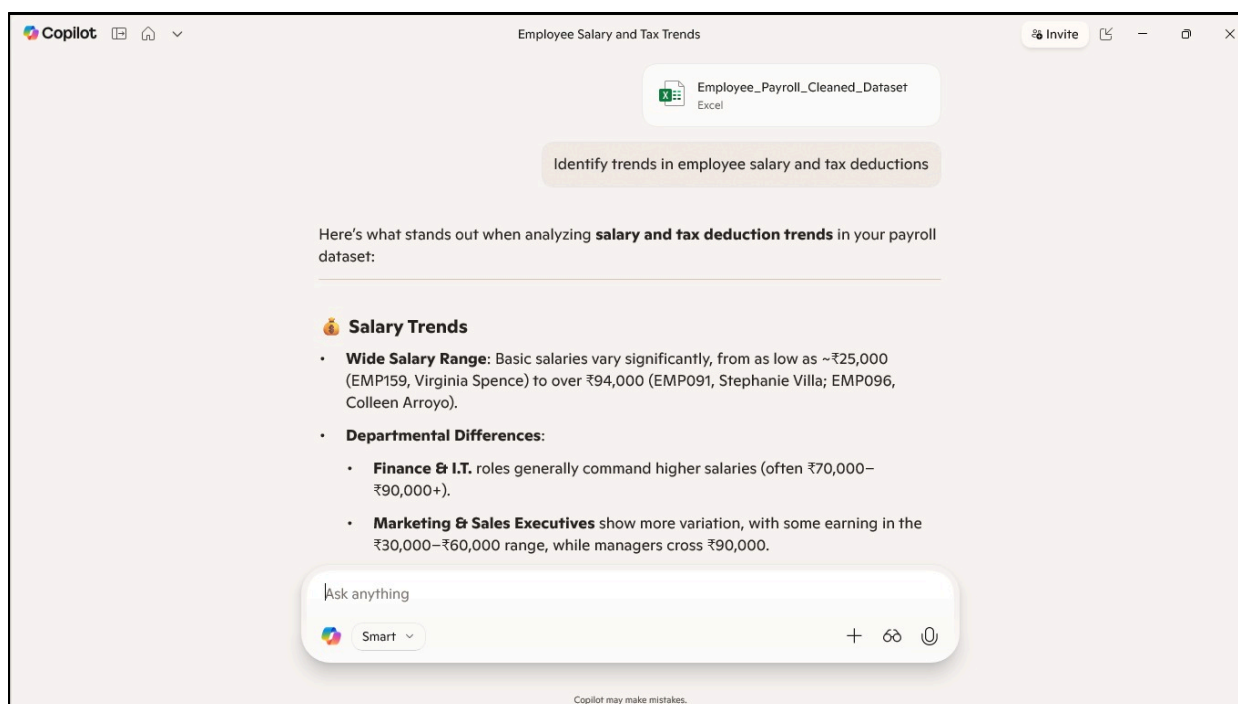


Figure : 6 Microsoft Copilot Integration with the project

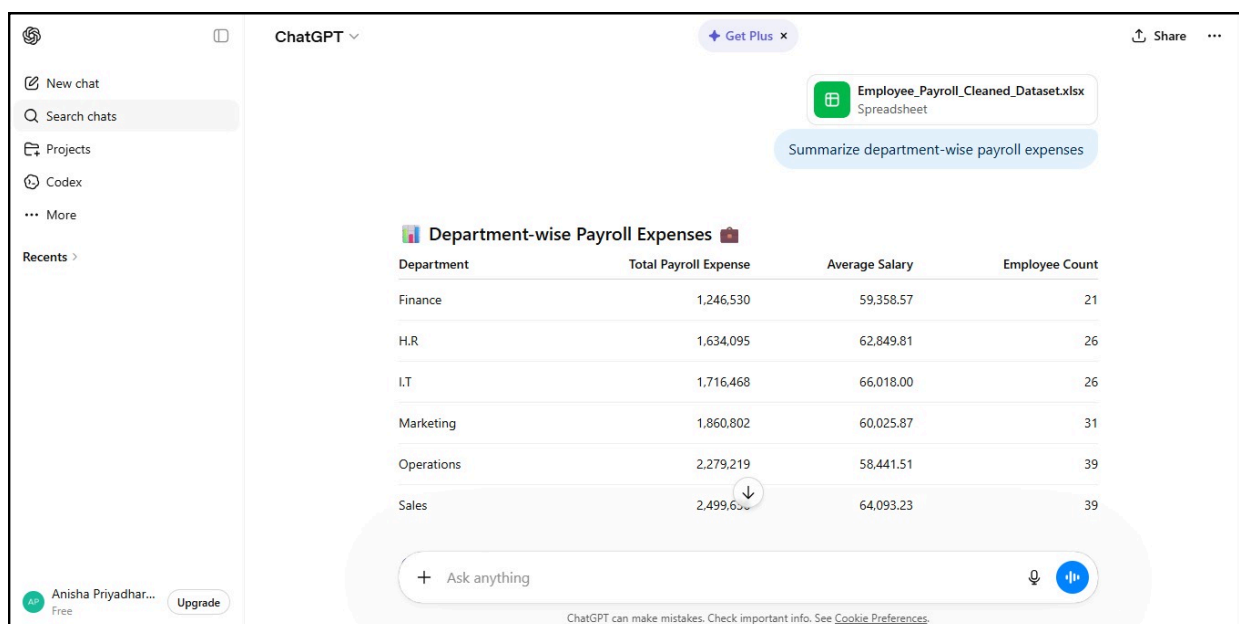


Figure : 7 Open AI ChatGPT Integration with the project

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Payroll data cleaning improved data accuracy and consistency
- Duplicate records and missing values were corrected successfully
- Financial analysis helped identify salary and tax patterns
- Department-wise payroll expenses were analysed effectively
- Charts and dashboards improved data visualization and understanding
- AI tools generated useful payroll summaries and insights
- AI integration reduced manual effort in reporting and analysis

5.2 Limitations

Although the project achieved its objectives, some limitations exist:

- The project uses a static payroll dataset
- Real-time payroll updates are not included
- Advanced machine learning models are not implemented
- The analysis is limited to Excel and AI-based tools

5.3 Project Link

GitHub Link of the Project [<https://github.com/anishapriyadharshini/Employee-Payroll-Data-Cleaning-And-Financial-Analysis-.git>]

5.4 Future Work

The project can be further enhanced in the following ways:

- Real-time payroll systems can be integrated
- Cloud database support can be added
- Advanced AI and machine learning models can be implemented
- Fraud detection and payroll prediction features can be added
- Interactive web-based dashboards can be developed

5.5 Conclusion

The project demonstrates the importance of payroll data cleaning and financial analysis in improving payroll accuracy and reporting efficiency. Microsoft Excel and AI tools were effectively used to clean payroll data, perform analysis, create visual dashboards, and generate meaningful insights.

The project helped in understanding how data analytics and Artificial Intelligence can support better payroll management and financial decision-making in organizations.

REFERENCES

List of all references, research papers, tutorials, and documentation used in project format below:-

1. **Microsoft Corporation**

Microsoft Excel Documentation <https://support.microsoft.com/excel>

2. **Microsoft Corporation**

Power BI Documentation <https://learn.microsoft.com/power-bi>

3. **Microsoft Corporation**

Microsoft Copilot – AI Assistant <https://copilot.microsoft.com>

4. **Edunet Foundation**

AICWFellowshipProgramGuidelines, 2026

5. **Kaggle Inc.**

Employee Payroll Dataset <https://www.kaggle.com>

6. **Tutorials and learning resources from:**

- YouTube (Power BI & Excel tutorials)
 - GitHub repositories on data analytics projects
-

APPENDICES

Appendix A - Sample Dataset

Employee_ID	Employee_Name	Gender	Age	Department	Designation	Joining_Date	Basic_Salary	Allowances	Bonus	Tax_Deductible	Net_Salary	Attendance_Percentage	Performance_Ratio	Payment_Date
EMP001	Bryan Davis	Male	29	Operations	Operations Executive	03-08-2019	₹ 81,395.00	₹ 10,834.00	₹ 11,533.00	₹ 11,959.27	₹ 91,805.73	85%	5.0	24-11-2026
EMP002	Michael Farley	Male	29	Sales	Sales Manager	01-01-2022	₹ 81,017.00	₹ 11,770.00	₹ 4,228.00	₹ 13,499.82	₹ 83,545.38	92%	2.0	27-02-2026
EMP003	Daniel Day	Female	31	Finance	Accountant	24-12-2019	₹ 59,032.00	₹ 13,905.00	₹ 3,102.00	₹ 4,084.77	₹ 72,574.23	78%	3.0	13-01-2026
EMP004	Glenn Alvarez	Female	25	H.R.	HR Executive	27-07-2023	₹ 64,657.00	₹ 14,838.00	₹ 4,003.00	₹ 12,119.73	₹ 71,378.27	88%	4.0	21-05-2026
EMP005	Shawn Sandoval	Female	24	Marketing	SEO Analyst	27-12-2021	₹ 46,087.00	₹ 8,134.00	₹ 6,419.00	₹ 5,458.15	₹ 55,181.85	95%	5.0	06-05-2026
EMP007	Jenna David MD	Female	38	H.R.	HR Manager	17-03-2020	₹ 72,709.00	₹ 3,725.00	₹ 5,085.00	₹ 7,230.49	₹ 74,288.52	81%	5.0	25-04-2026
EMP008	Briana Hamilton	Male	29	Sales	Sales Manager	06-06-2019	₹ 78,018.00	₹ 15,947.00	₹ 9,903.00	₹ 14,739.10	₹ 88,828.90	90%	3.0	13-03-2026
EMP009	Daniel Thompson	Male	40	I.T.	System Analyst	02-04-2023	₹ 77,529.00	₹ 8,134.00	₹ 12,759.00	₹ 14,780.75	₹ 91,380.25	79%	3.0	22-02-2026
EMP010	Brian Smith	Male	24	Operations	Operations Executive	05-01-2025	₹ 28,693.00	₹ 5,851.00	₹ 1,141.00	₹ 5,399.16	₹ 30,115.84	84%	4.0	13-12-2026
EMP011	Mary Burke	Male	24	I.T.	System Analyst	22-04-2025	₹ 63,295.00	₹ 11,730.00	₹ 9,747.00	₹ 10,097.65	₹ 74,704.35	97%	4.0	07-10-2026
EMP012	Gary Garcia	Female	23	Marketing	SEO Analyst	22-09-2022	₹ 89,317.00	₹ 11,548.00	₹ 11,378.00	₹ 12,027.22	₹ 1,00,216.78	89%	5.0	14-02-2026
EMP013	Robert Hernandez	Male	29	H.R.	HR Executive	27-10-2022	₹ 83,934.00	₹ 15,782.00	₹ 9,372.00	₹ 17,038.18	₹ 91,449.82	93%	3.0	30-10-2026
EMP015	Linda Graham	Male	36	Operations	Operations Executive	01-08-2019	₹ 67,423.00	₹ 11,560.00	₹ 4,074.00	₹ 8,022.36	₹ 75,034.64	74%	3.0	05-08-2026
EMP016	Ian Robbins	Female	35	Finance	Accountant	15-10-2020	₹ 82,597.00	₹ 5,830.00	₹ 1,448.00	₹ 11,082.28	₹ 78,792.72	86%	3.0	17-01-2026
EMP017	Danielle Doyle	Female	28	Sales	Sales Manager	07-07-2020	₹ 69,289.00	₹ 12,859.00	₹ 1,904.00	₹ 12,218.02	₹ 71,810.98	91%	4.0	16-01-2026
EMP018	Joshua Campbell	Female	21	I.T.	Software Engineer	13-10-2022	₹ 42,787.00	₹ 11,581.00	₹ 8,820.00	₹ 10,308.28	₹ 52,739.72	79%	3.0	17-02-2026
EMP019	Tara Price	Male	54	H.R.	HR Executive	16-04-2025	₹ 84,989.00	₹ 8,595.00	₹ 12,711.00	₹ 12,772.02	₹ 93,512.98	83%	5.0	22-04-2026
EMP020	Austin Reyes DDS	Male	28	Finance	Accountant	29-09-2022	₹ 61,611.00	₹ 10,187.00	₹ 14,799.00	₹ 6,910.58	₹ 75,883.42	96%	2.0	30-04-2026
EMP022	Mary Clark	Male	52	Operations	Operations Executive	01-10-2021	₹ 33,009.00	₹ 7,181.00	₹ 3,044.00	₹ 5,846.94	₹ 37,387.06	87%	5.0	16-09-2026
EMP023	Barbara Glover	Female	33	Finance	Finance Analyst	01-11-2020	₹ 73,329.00	₹ 15,846.00	₹ 5,815.00	₹ 16,374.99	₹ 78,615.01	80%	3.0	05-11-2026
EMP025	Andrew Garcia	Female	41	H.R.	HR Manager	07-01-2019	₹ 81,821.00	₹ 6,237.00	₹ 13,824.00	₹ 13,024.42	₹ 88,857.58	94%	5.0	14-01-2026
EMP026	Nicole Ford	Female	37	Finance	Finance Analyst	22-11-2023	₹ 72,018.00	₹ 8,134.00	₹ 6,280.00	₹ 6,053.86	₹ 87,067.14	82%	1.0	15-10-2026
EMP027	Brian Cook	Male	22	Finance	Accountant	25-12-2019	₹ 77,180.00	₹ 7,100.00	₹ 4,366.00	₹ 6,831.51	₹ 81,814.49	98%	3.0	12-04-2026
EMP028	Vanessa Bryant	Female	38	H.R.	HR Manager	13-09-2025	₹ 66,096.00	₹ 10,315.00	₹ 1,292.00	₹ 12,012.29	₹ 65,090.71	77%	3.0	29-11-2026
EMP029	Jason Marks	Male	39	I.T.	Software Engineer	27-03-2023	₹ 45,857.00	₹ 9,003.00	₹ 9,518.00	₹ 9,833.84	₹ 55,144.16	85%	1.0	26-12-2026
EMP030	David Manning	Female	32	I.T.	System Analyst	14-10-2019	₹ 60,251.00	₹ 18,097.00	₹ 10,319.00	₹ 20,984.69	₹ 95,992.31	92%	2.0	07-10-2026
EMP031	Jeremy Morris	Male	32	Operations	Operations Manager	28-01-2019	₹ 73,580.00	₹ 5,497.00	₹ 1,841.00	₹ 9,363.18	₹ 74,554.62	78%	1.0	03-08-2026
EMP032	Elizabeth Harper	Female	68	Sales	Sales Manager	16-07-2022	₹ 73,820.00	₹ 17,365.00	₹ 9,822.00	₹ 13,392.87	₹ 84,714.13	88%	5.0	23-04-2026
EMP033	Kallie Delacruz	Female	31	I.T.	System Analyst	11-01-2024	₹ 26,857.00	₹ 5,593.00	₹ 6,440.00	₹ 9,005.74	₹ 32,864.26	95%	3.0	18-08-2026
EMP034	Trevor Yang	Female	34	H.R.	HR Manager	22-02-2020	₹ 73,788.00	₹ 11,692.00	₹ 1,228.00	₹ 5,834.63	₹ 80,833.37	81%	4.0	29-10-2026
EMP035	Alexis Neal	Female	40	Sales	Sales Executive	29-06-2019	₹ 38,273.00	₹ 17,333.00	₹ 11,362.00	₹ 8,421.42	₹ 56,538.58	90%	3.0	23-05-2026
EMP036	Michael Williams	Female	28	Sales	Sales Manager	15-01-2020	₹ 82,438.00	₹ 3,934.00	₹ 2,028.00	₹ 11,179.73	₹ 77,218.27	76%	5.0	04-01-2026
EMP037	Lawrence Griffith	Female	28	H.R.	HR Manager	05-08-2020	₹ 77,738.00	₹ 12,845.00	₹ 9,298.00	₹ 17,258.98	₹ 82,418.02	84%	3.0	23-03-2026
EMP038	Amy Wolfe	Male	20	Sales	Sales Executive	01-06-2020	₹ 84,494.00	₹ 9,125.00	₹ 10,202.00	₹ 15,322.02	₹ 88,498.98	97%	5.0	24-12-2026

Figure : 8 Sample Dataset of Employee Payroll (Cleaned Version)

In Figure 8, I presented the cleaned Employee Payroll Dataset used for payroll data cleaning and financial analysis.

The dataset contains employee details such as Employee ID, Department, Designation, Salary Components, Tax Deduction, Net Salary, and Payment Date in a structured format. After preprocessing and standardization, the dataset became suitable for financial analysis, visualization, and dashboard creation.

Dataset link :

https://1drv.ms/x/c/815378ecef3556e1/IQDCnjKFxN0NQo_uZhG3bHkEAYdr6wWGY9lrQf0QZL9BQMcf?e=Zx8vM5

Appendix B - Key Excel Formulas / DAX Measures Used

Excel Formulas Used:

- SUM Formula – Used to calculate total salary, bonus, and payroll expenses
- AVERAGE Formula – Used to calculate average employee salary
- IF Formula – Used to apply conditions for salary and tax calculations
- COUNTIF Formula – Used to count duplicate or specific records in the dataset
- VLOOKUP Formula – Used to retrieve employee details from the dataset
- Pivot Tables – Used to summarize and analyse payroll data efficiently

Appendix C — AI Prompts Used

Some of the key prompts used in AI tools:

- “Summarize department-wise payroll expenses”
- “Identify salary and tax trends”
- “Generate payroll analysis summary”
- “Provide recommendations based on payroll data”

Appendix D — Screenshot Gallery

Screenshot : 1

The top screenshot shows the 'Employee_Payroll_Raw_Dataset' in Excel. The data is organized in columns A through Q, with rows 1 to 21. The columns include Employee_ID, Employee_Name, Gender, Age, Department, Designation, Joining_Date, Basic_Salary, Allowances, Bonus, Tax_Deduction, Net_Salary, Attendance_Percentage, Performance_Rating, and Payment_Date.

The bottom screenshot shows the 'Employee Payroll Dataset' in Excel. A dialog box is displayed, stating: "Excel 10 duplicate values found and removed; 200 unique values remain." The dialog box includes a link to "Give Feedback to Microsoft" and an "OK" button.

Figure : 9 Screenshots Of Removing Duplicates from Dataset

Screenshot : 2

The figure consists of three screenshots from an Excel spreadsheet, illustrating data cleaning steps.

Top Screenshot: Shows a list of employees with columns: Employee, Employee Name, Gender, Age, Department, Designation, Joining Date, Basic Salary, Allowance, Bonus, Tax Deduction, Net Salary, Performance Rating, and Payment Date. The data includes employees like Evan Daw, John Penr, Jennifer H, Stacy Bow, Nicholas C, Sandra Tr, Ryan Torr, Jeffrey Mil, Michael Ti, and Tonya Wol.

Middle Screenshot: Shows the 'Find and Replace' dialog box. The 'Find' field contains 'hr' and the 'Replace with' field contains 'H.R.'. The 'Find next' button is highlighted.

Bottom Screenshot: Shows the 'Allowances' tab with a detailed list of employees and their financial data. The 'More Number Formats...' menu is open, showing options like General, Number, Currency, Accounting, Short Date, Long Date, Time, Comma, Percentage, Fraction, Scientific, and Text.

Figure : 10 Screenshots Of other cleaning process of Dataset

Screenshot : 3

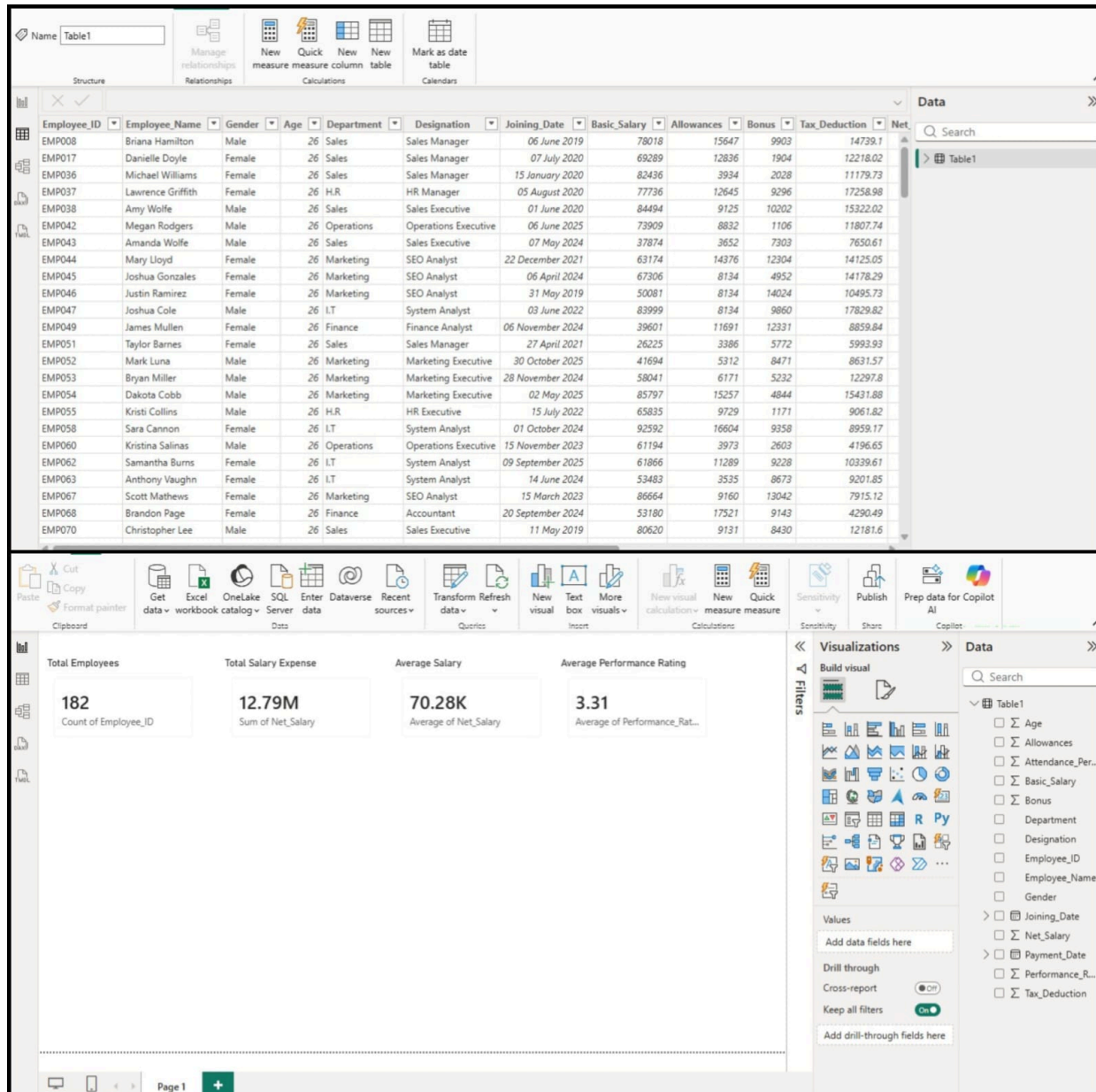


Figure : 11 Screenshots Of creating Cards for Dashboard using the Dataset

Screenshot : 4

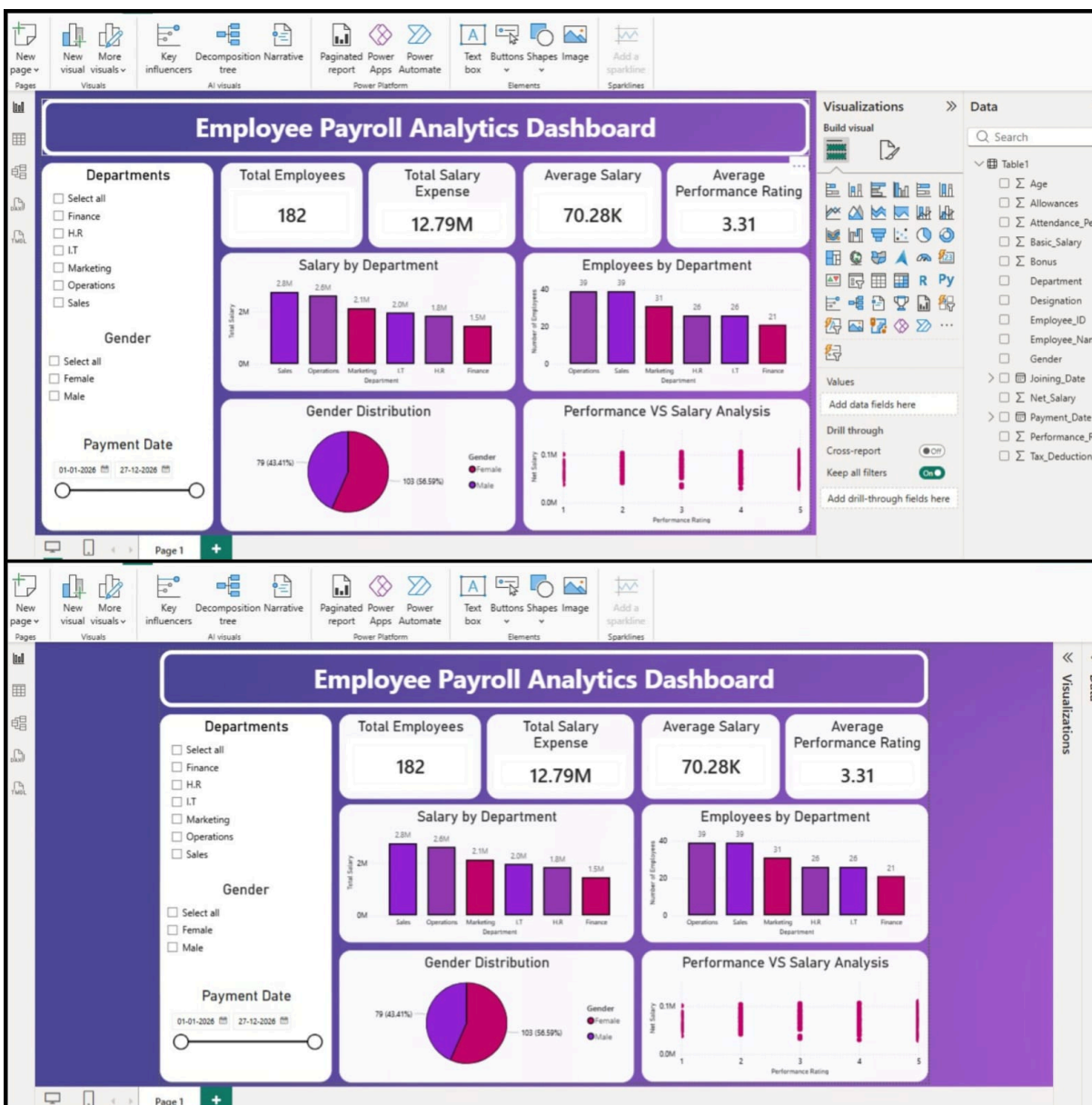


Figure : 12 Screenshots Of Formatting and Visualizing the Dashboard

Thank You